DM-2026-008

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TO: PSI Team, PIM, NASA JSC CUI Program Manager

SUBJECT: CUI Classification: DI-PSI-CDH-GDS Desk Instruction

OFFICE OF PRIMARY RESPONSIBILITY: Payload Software Integration (PSI)

1. Purpose

This memorandum documents the decision to classify the DI-PSI-CDH-GDS "C&DH / GDS / IT Security Coordination Meeting Guide" desk instruction as **CUI//SP-EXPT//FEDCON** in its as-shipped state, and to flag the potential applicability of the **ISVI (Information Systems Vulnerability Information)** category for NASA JSC CUI Program Manager adjudication.

2. Background

DI-PSI-CDH-GDS is a 25-page companion desk instruction for the Doc 8604 Rev D checklist. Unlike the blank checklist (which contains only empty form fields and publicly-citable SSP document numbers), the desk instruction contains **procedural knowledge about ISS information infrastructure**:

- **NTP architecture:** iPEHG-2 ISS/GPS time path and SSC NTP server ground-time path
- **Antivirus update distribution:** PL NAS, SSC System, and PD-uplink methods with access protocol specifics (SMB, NFS/AFP, HTTP URL)
- **KuIP protocol assignment rules:** Port-numbering constraints (>1023), APID encoding structure, NAT IP assignment procedures, Space/Ground Node ID management
- **JSL network topology:** Hirschmann Switch, WAPs, Edge Router connection types, static route handling, VLAN configuration guidance
- **HOSC GDS routing:** PDSS CCSDS Space Packet routing, iVODS paths, ICU on-board LOS storage, GSE packet assignment
- **D684-11372-01 identifier encoding:** APID bit-field structure, Subset ID encoding rules, PUI field definitions, KuIP PDL Log management procedures
- **MAC address assignment procedures:** Configurable vs. hardcoded determination, who assigns (PSI vs. PD), and how assignments flow to the Preliminary SW ICD

- **IT Security:** SSP 50974 requirements flow, malware-definition update paths, SSH key pair and X.509 certificate exchange procedures

This procedural knowledge describes *how the ISS information infrastructure operates*, even without disclosing specific IP addresses or payload configurations. An adversary with this knowledge would understand the architecture, data paths, assignment authority chains, and identifier structures of the ISS C&DH system.

3. Analysis

Export Controlled (SP-EXPT): The desk instruction describes technical data about the ISS payload C&DH and GDS systems that is subject to U.S. export control under the EAR (15 CFR Parts 730–774) and/or ITAR (22 CFR Parts 120–130). The ISS C&DH system architecture, including protocol assignment rules, identifier encoding, and network topology, constitutes controlled technical data about a space system. **SP-EXPT applies.**

Information Systems Vulnerability Information (ISVI): The CUI Registry defines ISVI as information about vulnerabilities in information systems. The desk instruction describes:

- Network topology (which devices connect where)
- Authentication paths (SSH key exchange, X.509 certificate procedures)
- Time-synchronization architecture (NTP sources)
- Malware-definition update distribution channels
- Identifier encoding that could aid spoofing (APID bit-field structure)

Whether this constitutes "vulnerability information" or merely "system description" is a judgment call that **requires NASA JSC CUI PM adjudication**. This decision conservatively marks the document as SP-EXPT and flags ISVI for determination.

Limited Dissemination Control — FEDCON: The desk instruction is used by Boeing PSI engineers (federal contractors) supporting NASA payloads. The ISS is an international-partner program (JAXA, ESA, CSA, Roscosmos), so NOFORN is inappropriate. FEDCON (Federal Employees and Contractors Only) is the correct LDC. If international-partner access is needed, the NASA JSC CUI PM may authorize REL TO markings on a case-by-case basis.

4. Decision

DI-PSI-CDH-GDS is classified as CUI//SP-EXPT//FEDCON.

- The desk instruction carries CUI//SP-EXPT//FEDCON banner on every page (running header and footer)
- The cover page explicitly states the classification and notes the ISVI-pending determination
- The desk instruction must be distributed only via NASA ORBIT Files (preferred) or encrypted email (DoD SAFE, NASA Box CUI, FIPS 140-2/3 S/MIME)

- The desk instruction must not be committed to any public repository
- The desk instruction must not be emailed unencrypted
- Destroy copies per NIST SP 800-88

Pending action: NASA JSC CUI PM to adjudicate whether ISVI additionally applies. If yes, the banner becomes **CUI//SP-EXPT/ISVI//FEDCON**. This is tracked as compliance finding F-12 in `docs/compliance/findings-and-recommendations.md`.

Distinction from the blank checklist: Per DM-2026-007, the blank Doc 8604 Rev D checklist form is NOT CUI because it contains only empty fields and publicly-citable document numbers. The desk instruction is CUI because it contains the *procedural answers* — the infrastructure knowledge that the blank form merely asks about.

5. Implementation

- CUI//SP-EXPT//FEDCON banner added to DI-PSI-CDH-GDS-Meeting-Guide.pdf running header/footer via `docs/di-pandoc-header.tex`
- Cover page classification block updated via `docs/di-pandoc-before-body.tex`
- ISVI-pending note included on cover page referencing compliance finding F-12
- Repository (`ISS_Payload_CDH_Interface_PSI_Checklist`) resides in a private Git repository — verified, no public-release violation
- The companion PSI Training Module spreadsheets (`PSI-TrainingModule_600-601*.xltx`, `PSI-TrainingModule_602*.xlsx`) are also CUI//SP-EXPT and co-reside in the same private repository

6. References

- 32 CFR Part 2002.4 (CUI definition): https://www.law.cornell.edu/cfr/text/32/part-2002
- 32 CFR Part 2002.20 (CUI marking requirements): https://www.law.cornell.edu/cfr/text/32/section-2002.20
- NARA CUI Registry — Export Controlled: https://www.archives.gov/cui/registry/category-detail/export-control
- NARA CUI Registry — ISVI: https://www.archives.gov/cui/registry/category-detail/information-systems-vulnerability-information
- NPR 2810.7 "Controlled Unclassified Information": https://nodis3.gsfc.nasa.gov/displayDir.cfm?t=NPR&c=2810&s=7
- `docs/compliance/findings-and-recommendations.md` — Finding F-12 (ISVI dual-category question)
- `docs/compliance/blank-form-cui-determination.md` — blank-form NOT CUI determination
- `docs/compliance/correctness-review.md` — citation verification

- DM-2026-006 (NASA CUI Directive Requirements Absorption)
- DM-2026-007 (Blank Checklist NOT CUI Determination)